



Chapter 2 Hydrostatics

Section 1 Hydrostatic pressure characteristics

Section 2 Stress equilibrium

Section 3 Pressure profile of a static fluid

Section 4 Manometer

Section 5 Hydrostatic pressure on a plane

Section 6 Hydrostatic pressure on a surface

Section 7 Buoyancy and floating-body stability

Chapter summary

Section 4 Manometer

- 
- 1, Piezometer tube
 - 2, U-shaped mercury manometer
 - 3, Differential pressure gauge
 - 4, Metal manometer
 - 5, Vacuum gauge



Section 4 Manometer

1, Piezometer tube

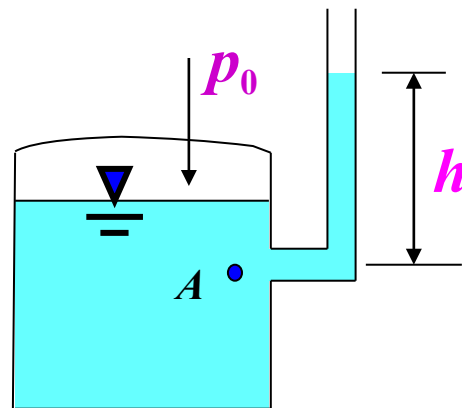
Piezometer tube: a vertical tube connecting the atmosphere to the point at which one wants to measure the static pressure.

Scope of application: The piezometer tube is suitable for measuring small pressures, but not suitable for measuring vacuum.

Calculated using principle of isobaric surface:

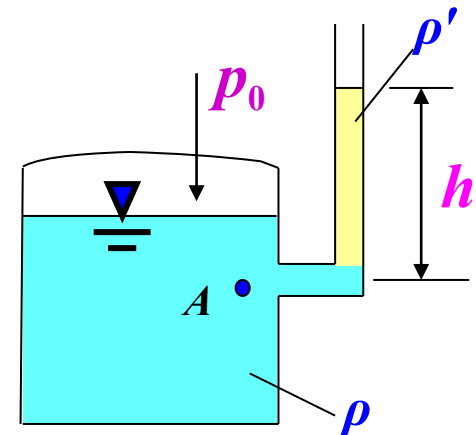
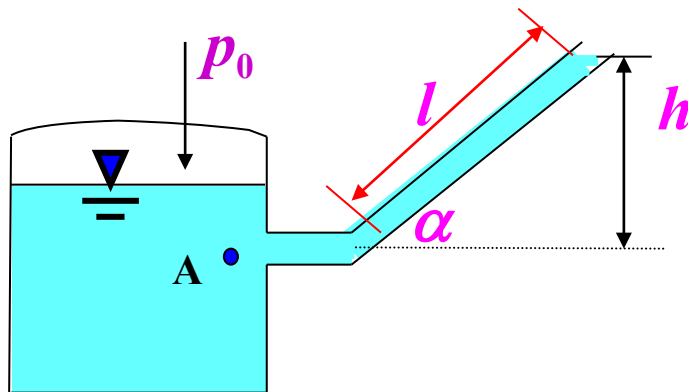
$$p_A = \rho gh$$

Question: Is the height of the liquid column in the piezometer tube the pressure head?



If the pressure of the measured point **A** is very small, in order to improve the measurement accuracy and increase the reading of the pressure measuring tube scale, the following two methods are often used:

(1) Place the piezometer tube obliquely, then the reading of the scale is l , and the pressure head is the vertical height h : $p_A = \rho gh = \rho g l \sin \alpha$.



(2) A light and water-insoluble liquid is placed in the piezometer tube, and the density $\rho' < \rho$ will have a larger h .





Section 4 Manometer

- 1, Piezometer tube
- 2, U-shaped mercury manometer
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2, U-shaped mercury manometer

Scope of application: It is used to measure the fluid pressure at a certain point in the pipeline or container where the pressure is usually large.

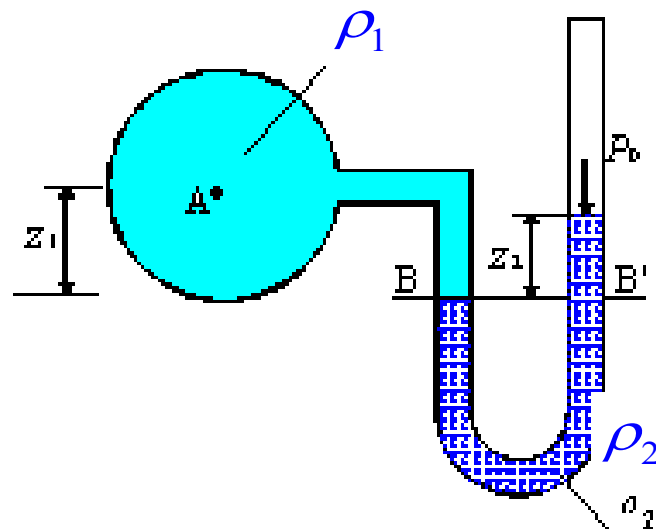
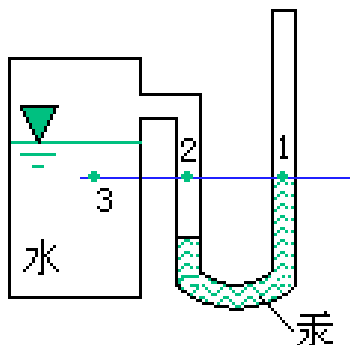
B-B' isobaric surface:

$$p_A + \rho_1 g z_1 = p_0 + \rho_2 g z_2$$

$$p_A = \rho_2 g z_2 - \rho_1 g z_1$$

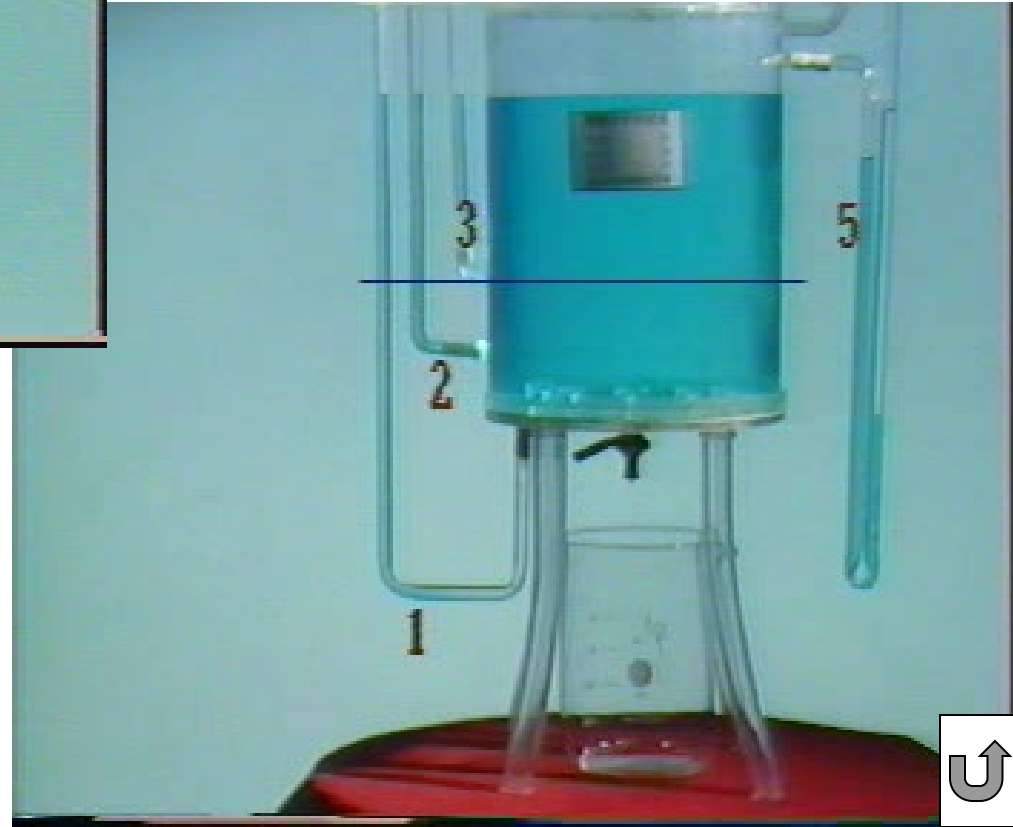
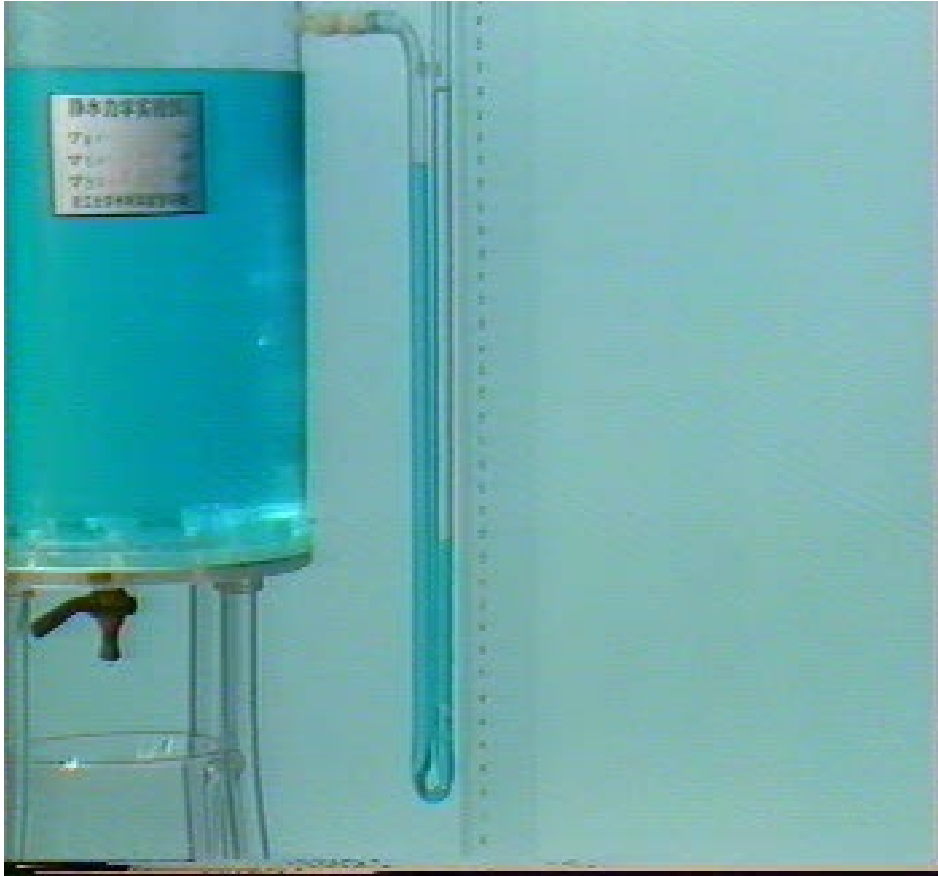
Choice:

- A. $p_1 = p_2 = p_3$;
- B. $p_1 > p_2 > p_3$;
- C. $p_1 < p_2 < p_3$;
- D. $p_2 < p_1 < p_3$.





U-shaped manometer





Section 4 Manometer

- 1, Piezometer tube
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3, Differential pressure gauge

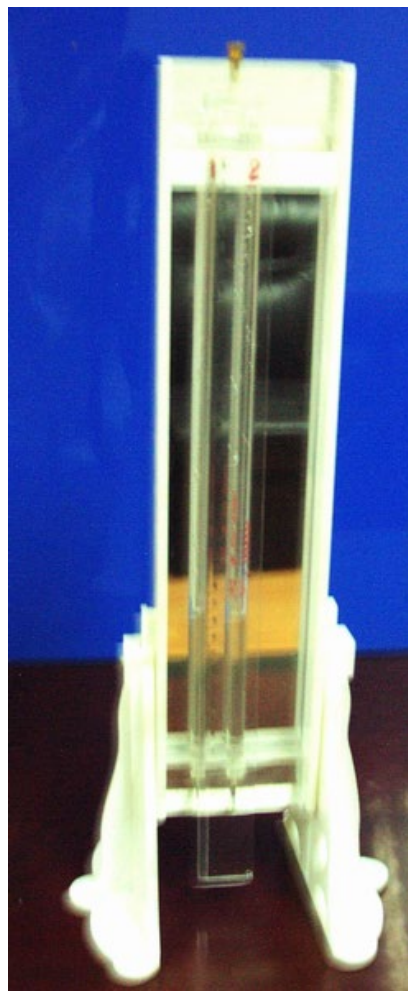
- Differential Pressure Gauge {
- Air differential pressure gauge: for medium and low pressure differentials
 - Oil differential pressure gauge: for low pressure differentials
 - Mercury differential pressure gauge: for high pressure differentials

Scope of application: measures the pressure difference between two points in the liquid or the hydraulic head difference.



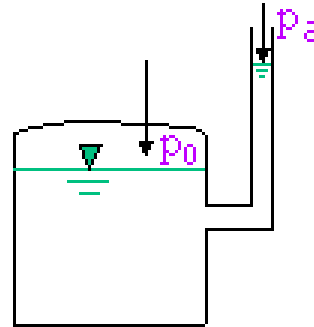


Differential pressure gauge



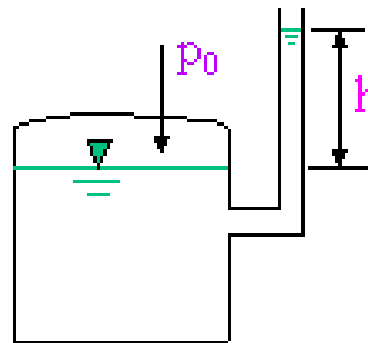
Choice:

- A. $p_0 = p_a$;
- B. $p_0 > p_a$;
- C. $p_0 < p_a$;
- D. can't judge.



Choice: $h=1.5\text{m}$, find the relative pressure p_0 of the liquid surface in units of height of the water column. The liquid in the container is gasoline. ($\rho g = 7.35 \text{ kN/m}^3$)

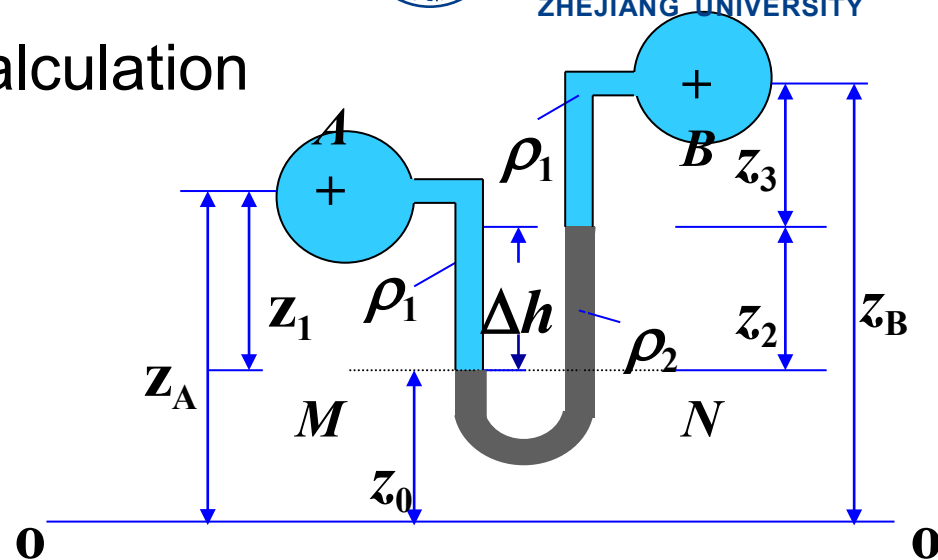
- A. 1.5 mH₂O;
- B. 1.125 mH₂O;
- C. 2 mH₂O;
- D. 11.5 mH₂O.



Differential Pressure Gauge Calculation

$$p_A - p_B$$

$$= \rho_1 g z_3 + \rho_2 g z_2 - \rho_1 g z_1$$



Top fluid = water, Bottom fluid = mercury, $z_2 = \Delta h$

$$\frac{p_A}{(\rho g)_w} - \frac{p_B}{(\rho g)_w} = z_3 + \frac{(\rho g)_{Hg}}{(\rho g)_w} z_2 - z_1 + z_0 - z_0 + z_2 - z_2$$

$\downarrow \Delta h$

$$(z_A + \frac{p_A}{(\rho g)_w}) - (z_B + \frac{p_B}{(\rho g)_w}) = (\frac{(\rho g)_{Hg}}{(\rho g)_w} - 1) \Delta h = 12.6 \Delta h$$

Conclusion: The hydraulic head difference is proportional to the displacement of the fluids.





Section 4 Manometer

- 1, Piezometer tube
- 2, U-shaped mercury manometer
- 3, Differential pressure gauge
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- 5, Vacuum gauge





4, Metal manometer

Scope of application: used to measure large pressure.
It is the most commonly used pressure measuring instrument in water works and pipeline systems.

5, Vacuum gauge

Scope of application: for measuring vacuum.



1. In traditional experiments, why is mercury often used as the working fluid of the U-shaped piezometer?

1. Low compressibility; 2. Low vaporization pressure; 3. High density.

2. The two points **A** and **B** are located in the same water tank. What is the liquid level difference $h_{\text{汞}}$ of the U-shaped mercury differential pressure gauge connecting the two points?

(1) $\frac{p_A - p_B}{(\rho g)_m}$

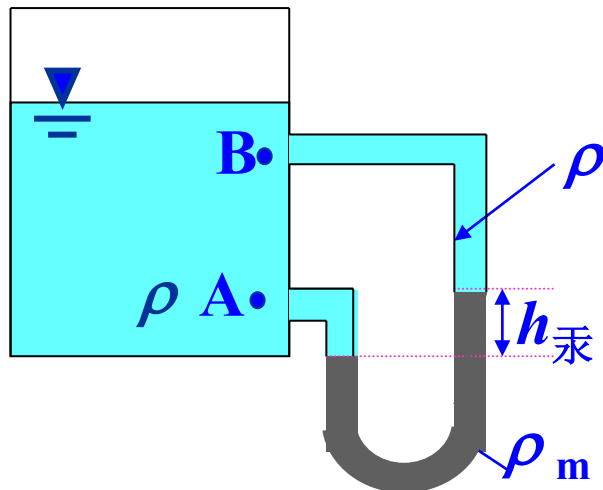
(2) $\frac{p_A - p_B}{(\rho g)_m - (\rho g)}$

(3) 0

Answer: (3)

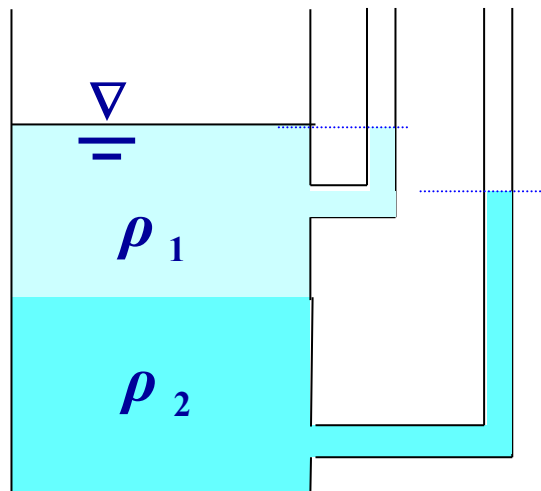
$$\Delta h_{\text{汞}} * 12.6 = (z_A + \frac{p_A}{\rho g}) - (z_B + \frac{p_B}{\rho g}) = 0$$

$$\therefore \Delta h_{\text{汞}} = 0$$



3. The two liquids shown in the figure are in the same container, and $(\rho g)_1 < (\rho g)_2$. Two pressure measuring tubes are installed on the side wall of the container. Are the surface levels of the liquids correct (qualitatively)?

Yes.





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Chapter summary





Hydrostatic pressure on a plane



1, Hydrostatic pressure diagram

2, Hydrostatic pressure on a plane

(1) Analytical method

(2) Graphical method



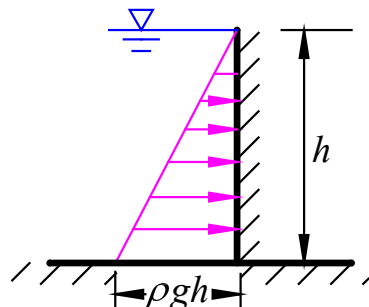
Hydrostatic pressure on a plane

1, Hydrostatic pressure diagram

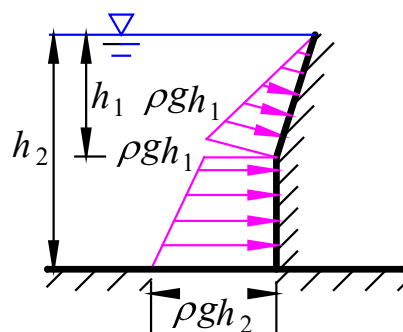
1, Size (scaled): $p = \rho gh$;

2, Direction: perpendicular to the surface; indicated by an arrow.

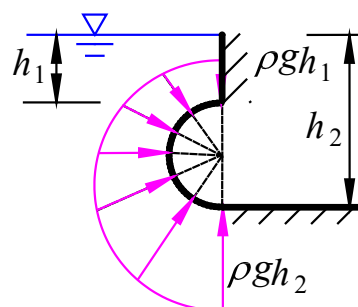
3, The outer line of the pressure distribution diagram: plane - straight line; curved surface - curve.



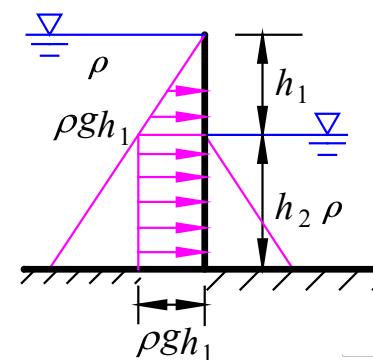
(a)



(b)



(c)



(d)





Hydrostatic pressure on a plane

1, Hydrostatic pressure diagram

2, Hydrostatic pressure on a plane

(1) Analytical method

(2) Graphical method



2, Hydrostatic force on a plane surface

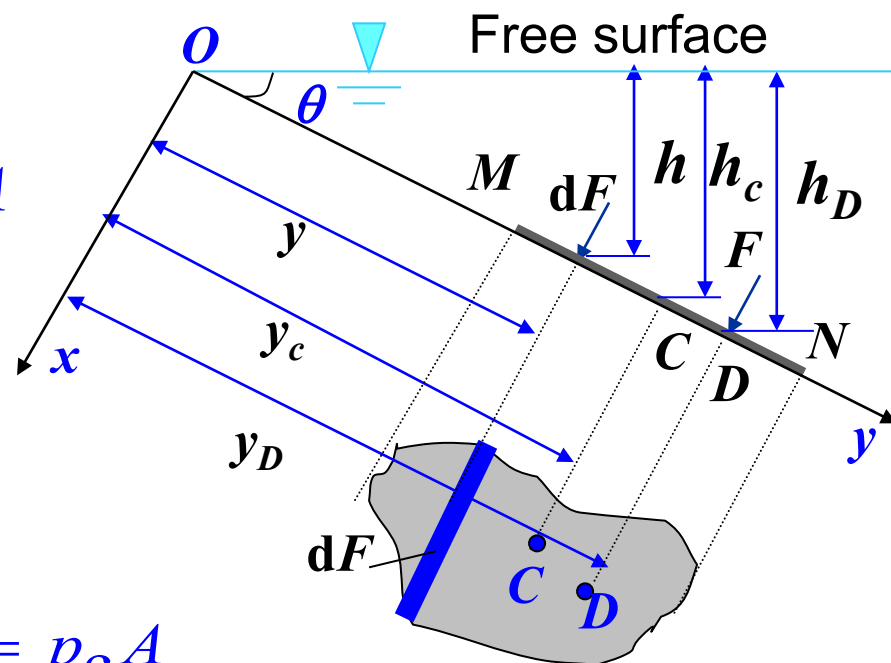
(1) Analytical method

1, Force on the infinitesimal area dA :

$$dF = p dA = \rho g h dA = \rho g y \sin \theta dA$$

Geometric center: $\int_A y dA = y_c A$

$$\begin{aligned} F &= \int dF = \int \rho g y \sin \theta dA \\ &= \rho g \sin \theta y_c \cdot A = \rho g h_c A = p_c A \end{aligned}$$



Conclusion: The hydrostatic force F on an arbitrarily-shaped plane submerged in fluid is equal to the plane's area A times the static pressure p_c at its geometric center.

2, Center of pressure

$$F = \rho g \sin \theta y_c \cdot A$$

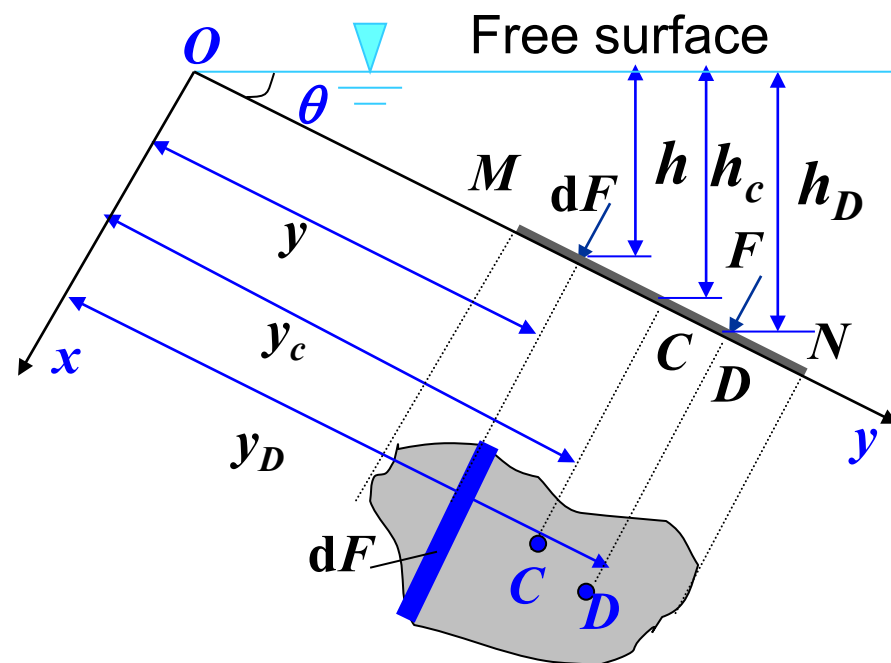
Torque on the **Ox**-axis:

$$F \cdot y_D = \int y \cdot dF = \rho g \sin \theta \int_A y^2 dA$$

Area moment of inertia:

$$\int_A y^2 dA = I_x = I_{cx} + y_c^2 A$$

$$y_D = \frac{\rho g \sin \theta \cdot I_x}{F} = \frac{I_x}{y_c A} = y_c + \frac{I_{cx}}{y_c A}$$

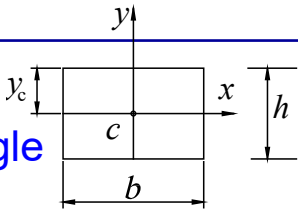
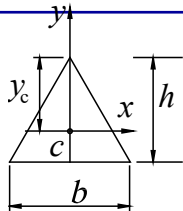
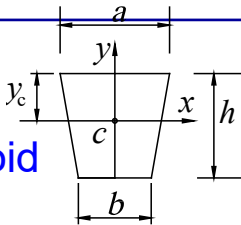
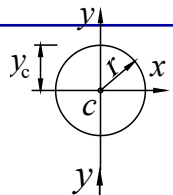
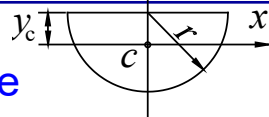


I_x is the moment of inertia of area **A** around the **Ox**-axis.

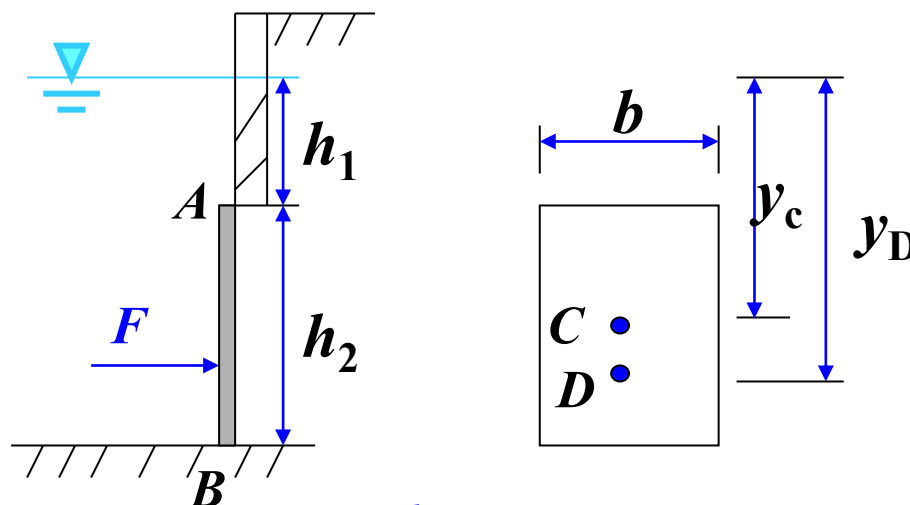
I_{cx} - Moment of inertia of area **A** around its centroid axis parallel to the **Ox**-axis.



A , y_c , and I_{cx} -values of common geometries

Geometry	Area A	Geometric center y_c	Moment of inertia through centroid axis I_{cx}
Rectangle 	bh	$\frac{1}{2}h$	$\frac{1}{12}bh^3$
Triangle 	$\frac{1}{2}bh$	$\frac{2}{3}h$	$\frac{1}{36}bh^3$
Trapezoid 	$\frac{1}{2}h(a+b)$	$\frac{h}{3}(\frac{a+2b}{a+b})$	$\frac{1}{36}h^3(\frac{a^2+4ab+b^2}{a+b})$
Circle 	πr^2	r	$\frac{1}{4}\pi r^4$
Semicircle 	$\frac{1}{2}\pi r^2$	$\frac{4}{3}\frac{r}{\pi}$	$\frac{9\pi^2 - 64}{72\pi}r^4$

Example 1 A vertical rectangular gate:
given $h_1=1\text{m}$, $h_2=2\text{m}$, and width $b=1.5\text{m}$,
find the total force and the **center of pressure**.



Solution:

$$F = p_c A = \rho g h_c A = \rho g \left(h_1 + \frac{h_2}{2} \right) \cdot b h_2$$

$$= 9800 \times \left(1 + \frac{2}{2} \right) \times 1.5 \times 2 = 58800 \text{ N} = 58.8 \text{ kN}$$

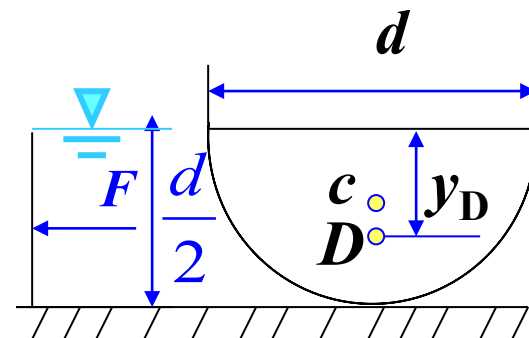
$$y_D = y_c + \frac{I_{cx}}{y_c A} = \left(h_1 + \frac{h_2}{2} \right) + \frac{\frac{1}{12} b h_2^3}{\left(h_1 + \frac{h_2}{2} \right) \cdot b h_2} = 2 + \frac{\frac{1}{12} \times 1.5 \times 2^3}{2 \times 1.5 \times 2} = 2.17 \text{ m}$$

Example 2 There is a vertical semicircular wall whose diameter is located on the liquid surface. Find the value of $\mathbf{F}$ and the center of pressure. ($I_x = \frac{\pi d^4}{128}$)

$$F = \rho g h_c A = p_c A$$

$$\begin{aligned} F &= p_c A = \rho g h_c A \\ &= \rho g \cdot \frac{4d}{6\pi} \cdot \frac{1}{8} \pi d^2 = \rho g \cdot \frac{1}{12} d^3 \end{aligned}$$

$$y_D = \frac{I_x}{y_c A} = \frac{\frac{\pi d^4}{128}}{\frac{4d}{6\pi} \cdot \frac{1}{8} \pi d^2} = \frac{3}{32} \pi d$$





Hydrostatic pressure on a plane

- 1, Hydrostatic pressure diagram
- 2, Hydrostatic pressure on a plane
 - (1) Analytical method
 - (2) Graphical method

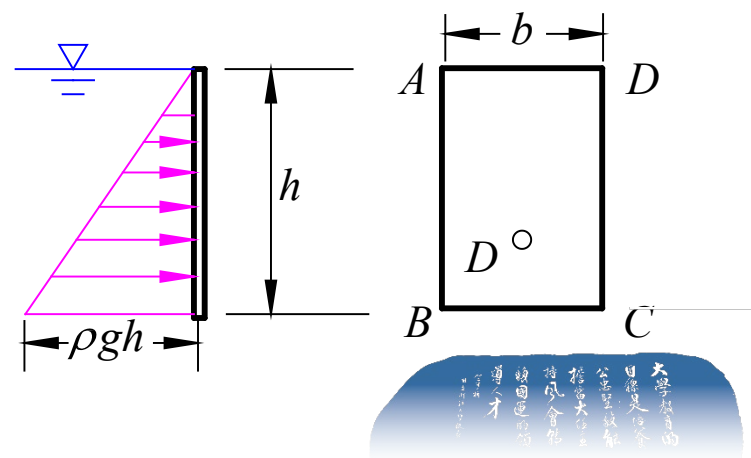


(2) Graphical method

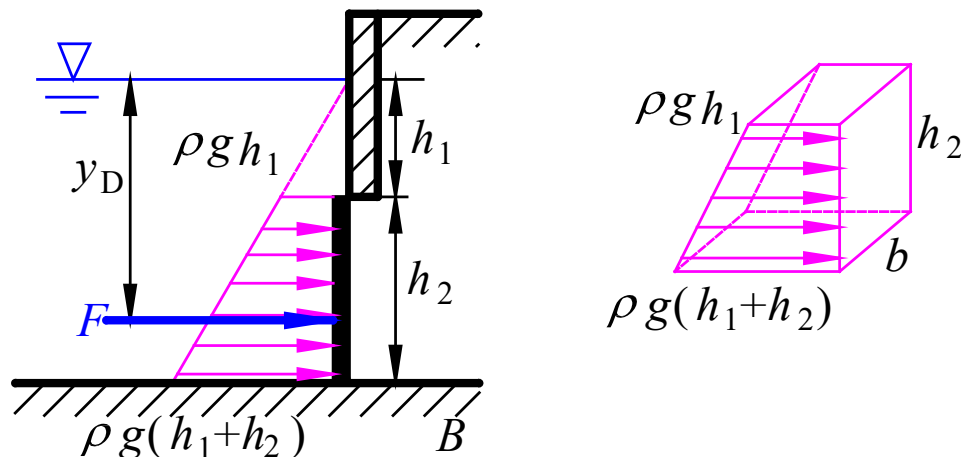
Scope of application: The total hydrostatic force and the center of pressure on a plane

Principle: The total hydrostatic force is equal to the **volume** of the pressure distribution diagram, and the center of pressure is the distance of its **geometric center D** from the free surface.

$$F = \frac{1}{2} \rho g h^2 \cdot b = \Omega \cdot b$$



Example 1 A vertical rectangular gate, given $h_1=1\text{m}$, $h_2=2\text{m}$, width $b=1.5\text{m}$, find the total force and the center of pressure. (same as Example 1 for analytical method)



The total force is the volume of the pressure diagram:

$$F = \frac{1}{2} [\rho g h_1 + \rho g (h_1 + h_2)] \cdot b \cdot h_2 = 58.8 \text{ kN}$$

The center of pressure is the distance of its geometric center from the surface:

$$y_D = h_1 + \frac{h_2}{3} \frac{\rho g h_1 + 2 \rho g (h_1 + h_2)}{\rho g h_1 + \rho g (h_1 + h_2)} = 2.17 \text{ m}$$



Example 2: Given a rectangular plane $h=1\text{m}$, $H=3\text{m}$, $b=5\text{m}$, find F and y_D .

Analytical method:

Length of the plane: $l = \frac{H-h}{\sin 30^\circ} = \frac{(3-1)\text{ m}}{0.5} = 4\text{ m}$

$$h_c = h + \frac{H-h}{2} = 1\text{ m} + \frac{(3-1)\text{ m}}{2} = 2\text{ m}$$

$$A = lb = (4\text{ m}) \times (5\text{ m}) = 20\text{ m}^2$$

$$F = \rho g h_c \cdot A = 9800 \times 2 \times 20 = 392\text{ kN}$$

$$y_D = y_c + \frac{I_{cx}}{y_c A}$$

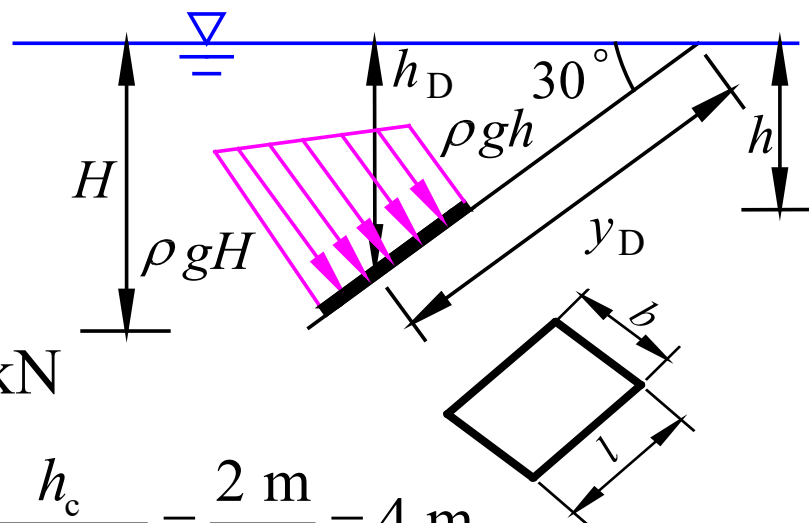
$$y_c = \frac{h_c}{\sin 30^\circ} = \frac{2\text{ m}}{1/2} = 4\text{ m}$$

$$I_{cx} = \frac{1}{12} b l^3 = \frac{1}{12} \times (5\text{ m}) \times (4\text{ m})^3 = \frac{80}{3}\text{ m}^4$$

Hence:

$$y_D = 4\text{ m} + \frac{80/3\text{ m}^4}{(4\text{ m}) \times (20\text{ m}^2)} = \frac{13}{3}\text{ m}$$

$$h_D = y_D \sin 30^\circ = \frac{13}{6}\text{ m}$$



Graphical method:

The pressure map is divided into two parts (triangle + rectangle)

$$\Omega = \frac{\rho g}{2} (H + h) \times \frac{H - h}{\sin 30^\circ} = 8\rho g$$

$$F = \Omega b = 8\rho g \times 5 = 40\rho g = 392\text{KN}$$

$$F_1 = \frac{\rho g}{2} (H - h) lb = 20\rho g$$

$$F_2 = \rho g h b l = 20\rho g$$

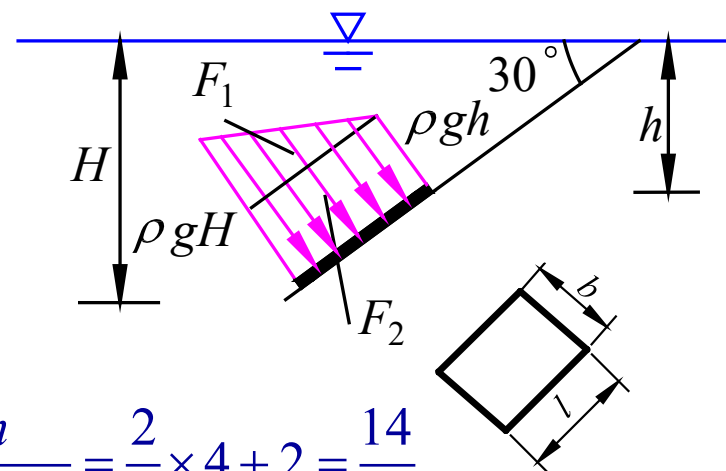
$$y_{D_1} = \frac{2}{3}l + \frac{h}{\sin 30^\circ} = \frac{2}{3} \times 4 + 2 = \frac{14}{3}$$

$$y_{D_2} = \frac{1}{2}l + \frac{h}{\sin 30^\circ} = \frac{1}{2} \times 4 + 2 = 4$$

$$\therefore F \cdot y_D = F_1 y_{D_1} + F_2 y_{D_2}$$

$$\therefore y_D = \frac{1}{2} (y_{D_1} + y_{D_2}) = \left(\frac{14}{3} + 4 \right) \times \frac{1}{2} = \frac{13}{3} \text{ m}$$

$$h_D = y_D \cdot \sin 30^\circ = \frac{13}{6} \text{ m}$$



Example 3 As shown in the figure, the left side is the water tank, the reading of the pressure gauge on it is $-0.147 \times 10^5 \text{ Pa}$, and it is separated by a gate with a width of 1.2 m , and the gate is hinged at point **A**. How much horizontal force **F** must be applied at point **B** in order for gate **AB** to be in equilibrium?

Graphical method

$$\therefore p_0 = \frac{-0.147 \times 10^5}{9800} = -1.5 \text{ mH}_2\text{O}$$

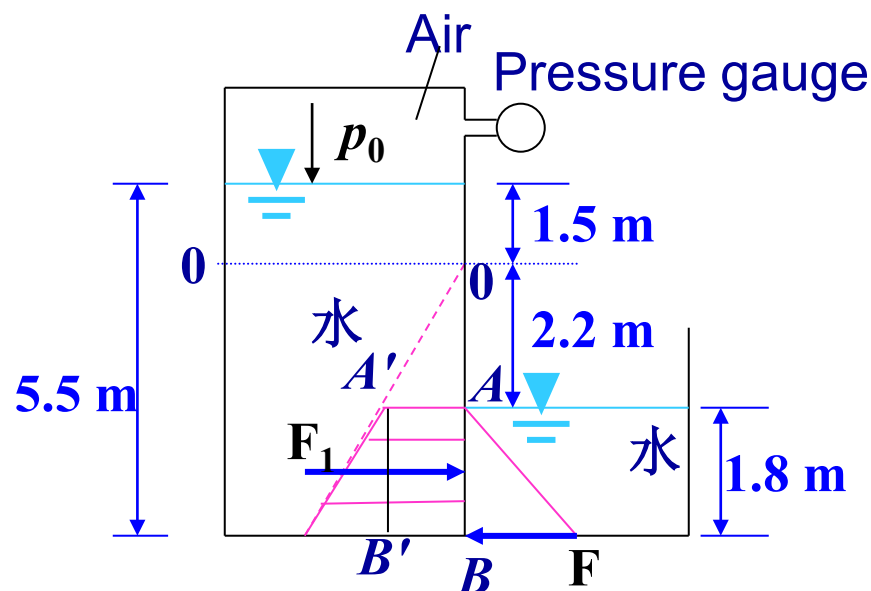
Hence, **0-0** is the zero-pressure surface

$$\begin{aligned} \therefore F_1 &= \Omega_{A'ABB'} \cdot b = 9.8 \times 2.2 \times 1.8 \times 1.2 \\ &= 46.57 \text{ kN} \end{aligned}$$

$$\sum M_A = 0$$

$$F_1 \times 0.9 - F \times 1.8 = 0$$

$$F = 23.29 \text{ kN (in left direction)}$$





Summary

1, Graphical method

Force magnitude: $F = \Omega \cdot b$

Pressure center: geometric center of pressure distribution diagram

2, Analytical method

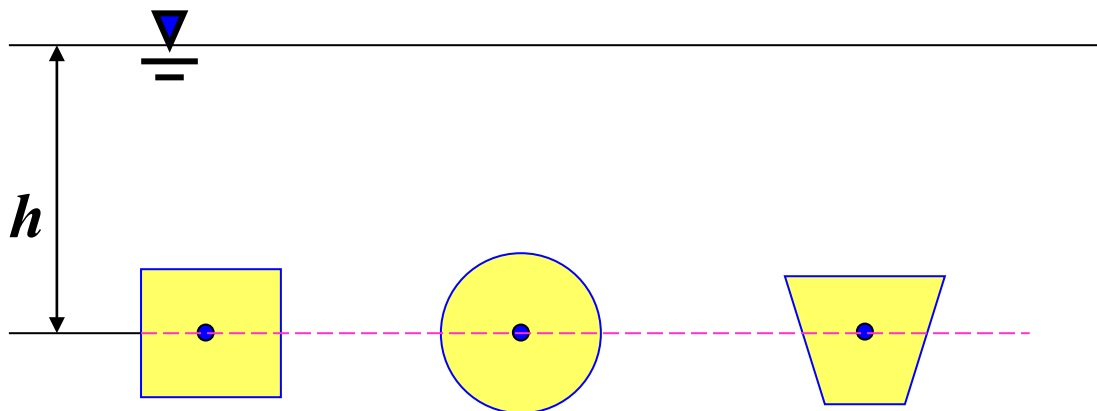
(find geometric center **C** of plane)

Force magnitude: $F = p_C \cdot A = \rho g h_c \cdot A$

Pressure center: $y_D = y_C + \frac{I_{cx}}{y_C \cdot A}$



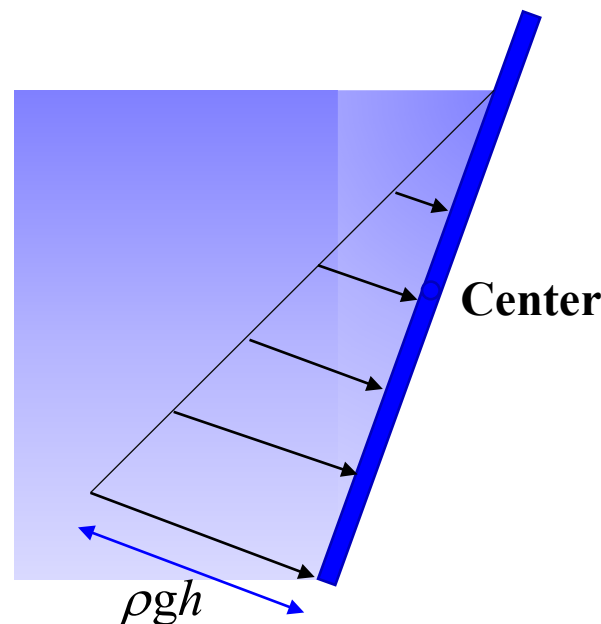
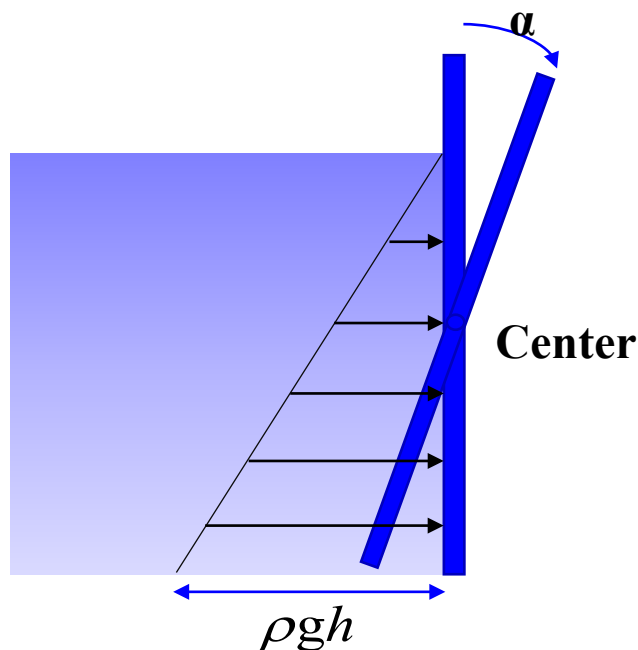
1. Three planes of different shapes immersed in water have the same area and the same depth of water at the centroid. Q: Which is subject to the greatest total hydrostatic force? Are the water depths at the pressure centers the same?



the same; not the same



2. A plane gate with a water-retaining area **A**, one side retains water, if it passes through its centroid **C**. If the horizontal axis rotates at any angle of α , does the magnitude and direction of the total hydrostatic force and the pressure center change?

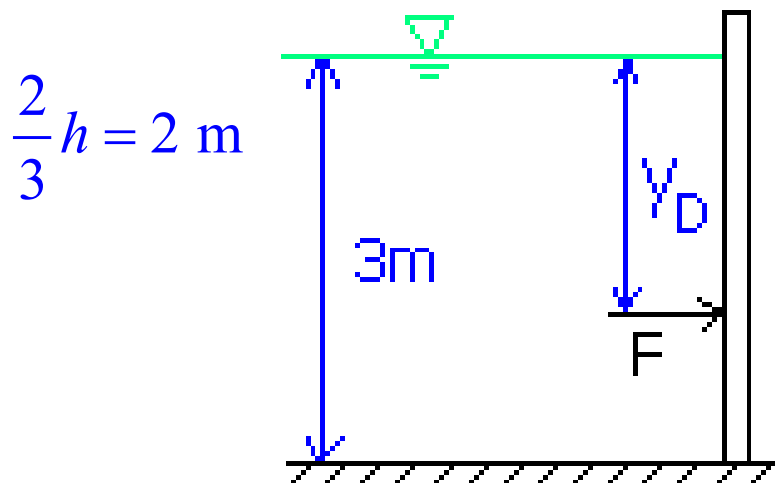


The size does not change; the direction changes, but it is always perpendicular to the gate; the center of pressure changes

3. For the graphical and analytical method, are there any restrictions on the shape of the submerged plane? Why?

Graphical method requires regular shape in order to draw the pressure distribution map. Analytical method has no restrictions.

4. A rectangular plate placed vertically to block water, the water depth is **3m**. What is the distance y_D between the center of pressure and the water surface?





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Chapter summary



Section 6 Hydrostatic pressure on a surface



1, Horizontal force

2, Vertical force

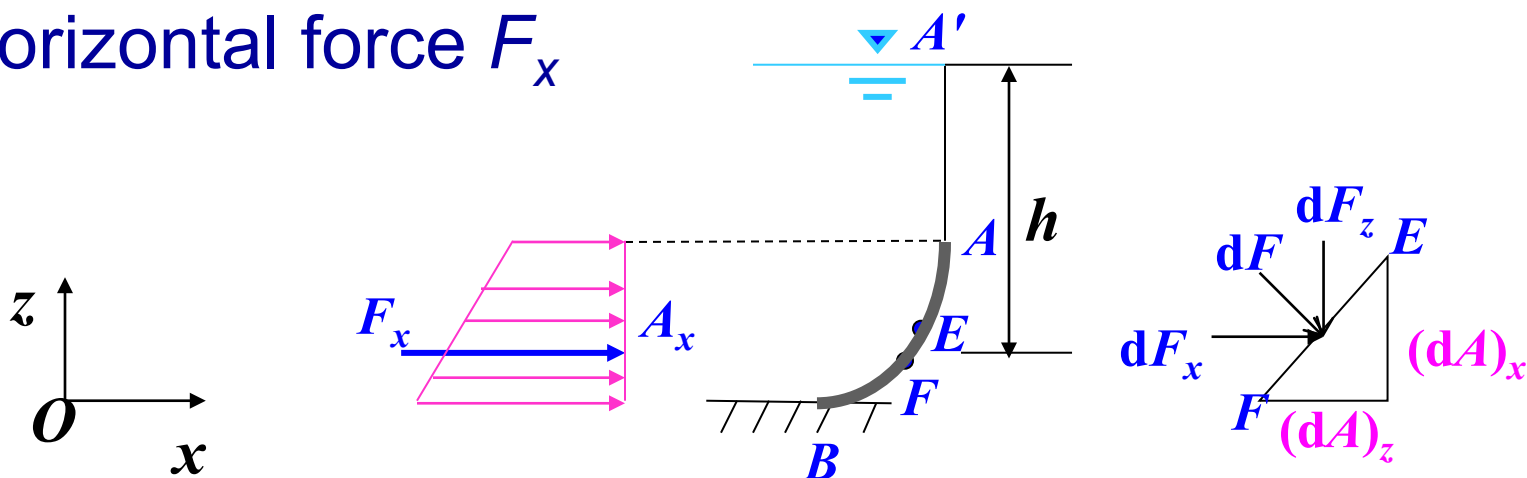
3, Pressure body

4, Total force



Section 6 Hydrostatic pressure on a surface

1, Horizontal force F_x



$$F_x = \int dF_x = \int_{A_x} \rho g h dA_x = \rho g h_c A_x$$

Conclusion: The horizontal component F_x of the total hydrostatic pressure F acting on a curved surface is equal to the total hydrostatic force acting on the vertical projection surface (rectangular plane) of the curved surface.



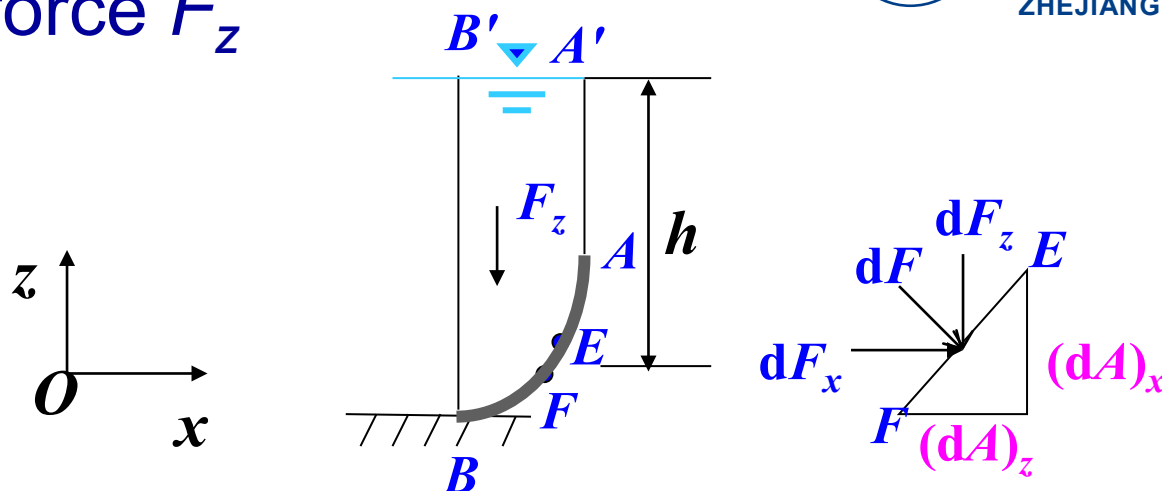


Section 6 Hydrostatic pressure on a surface

- 1, Horizontal force
- 2, Vertical force
- 3, Pressure body
- 4, Total force



2, Vertical force F_z



$$F_z = \int_{A_z} \rho g h dA_z = \rho g \int_{A_z} h dA_z = \rho g V_{ABB'A'} = \rho g V_p$$

V_p ——Volume of pressure body

Conclusion: The vertical component F_z of the total hydrostatic force $\mathbf{F}$ acting on the curved surface is equal to the weight of the liquid contained in the pressure body on the curved surface.



Section 6 Hydrostatic pressure on a surface

- 1, Horizontal force
- 2, Vertical force
- 3, Pressure body
- 4, Total force

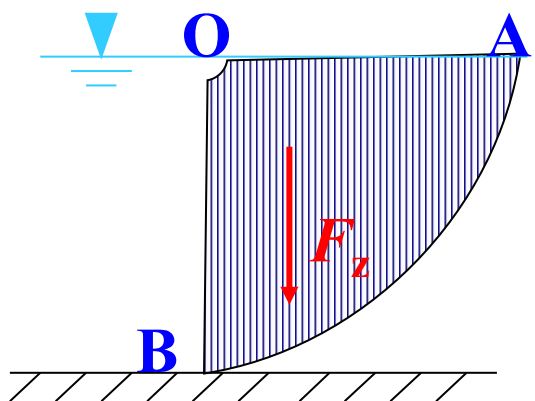


3, Pressure body

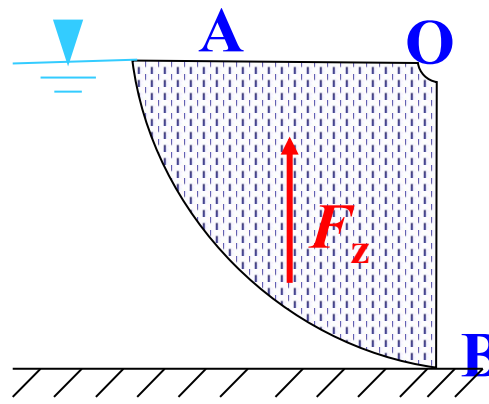
Composition of the pressure body volume:

- (1) The submerged surface itself;
- (2) the vertical plane made by the edge around the curved surface;
- (3) the free surface or extension of free surface

Types of pressure bodies: real pressure bodies and virtual pressure bodies.
The direction of the real pressure body F_z is downward, and the direction of the virtual pressure body F_z is upward.



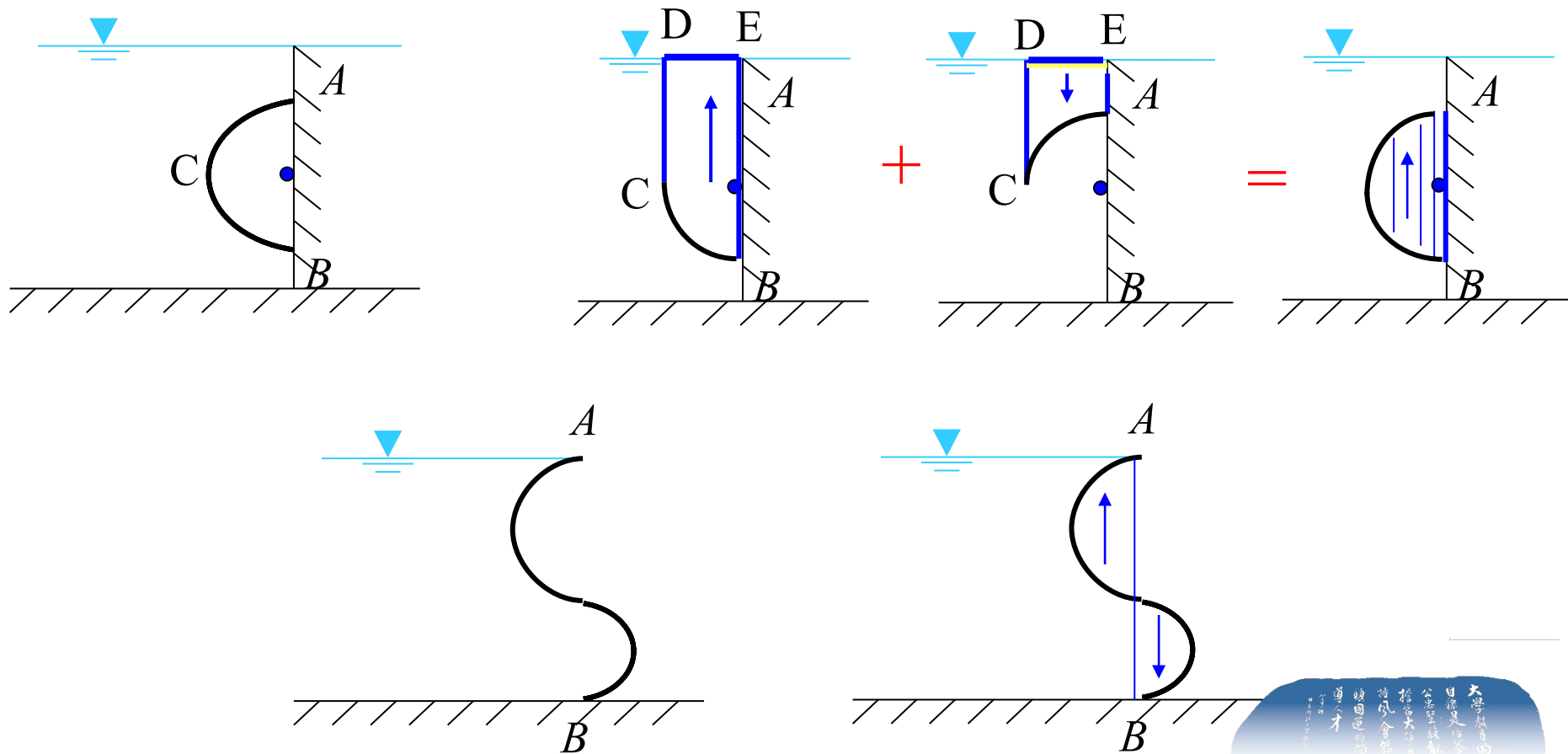
(a) real pressure body



(b) virtual pressure body



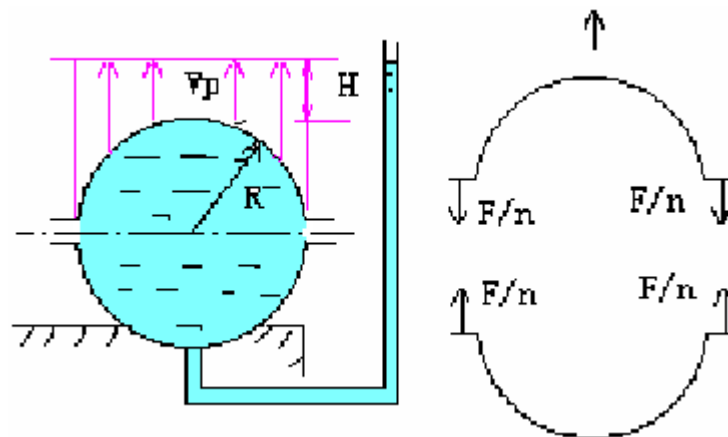
- Draw the pressure body on the **AB** surface in the drawing



Example 1 A spherical container is formed by riveting two hemispherical surfaces, there are n rivets, and the liquid with a weight of ρg is contained in it. Find the tensile force on each rivet.

Solution: Take the upper hemisphere of the spherical container as the curved surface, then the pressure body it is subjected to is shown in the figure. Then:

$$\begin{aligned}
 nF &= \rho g V_p \\
 &= \rho g \left[\pi R^2 (R + H) - \frac{2}{3} \pi R^3 \right] \\
 &= \rho g \left(\frac{1}{3} \pi R^3 + \pi R^2 H \right) \\
 \therefore F &= \frac{\rho g}{n} \left(\frac{1}{3} \pi R^3 + \pi R^2 H \right)
 \end{aligned}$$



Example 2 Use a steel plate with an allowable stress $[\sigma]$ **150MPa** to make a water pipe with a diameter **D** of **1m**. The pressure in the water

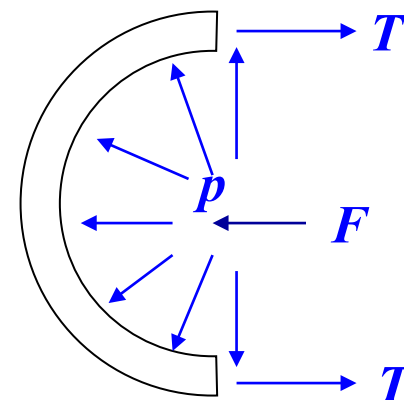
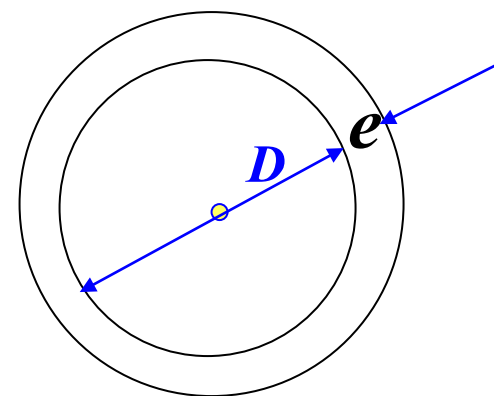
pipe is as high as under a **500m** water column. Find the thickness of the water pipe wall. (ignoring the pressure differences caused by the different heights of each point in the pipe, $e \ll D$)

Take length of pipe as **1m**: $F = p \cdot A_z = p \cdot D \cdot 1$,

$$2T = F = pD$$

$$T \leq [\sigma] \cdot e \cdot 1$$

$$\therefore e \geq \frac{T}{[\sigma]} = \frac{pD/2}{[\sigma]} = \frac{9800 \cdot 500 \cdot 1}{2 \cdot 150 \cdot 10^6} = 0.0163 \text{ m}$$





Section 6 Hydrostatic pressure on a surface

- 1, Horizontal force
- 2, Vertical force
- 3, Pressure body
- 4, Total force



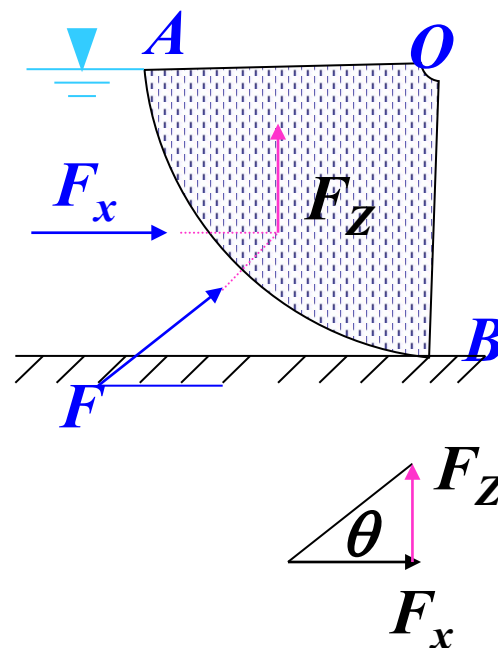
4, Total force

Total hydrostatic force acting on the surface

$$F = \sqrt{F_x^2 + F_z^2}$$

The angle between $\mathbf{F}$ and the horizontal plane:

$$\theta = \tan^{-1}\left(\frac{F_z}{F_x}\right)$$



Draw a line with angle θ through the intersection of the direction lines of $\mathbf{F}_x$ and $\mathbf{F}_z$. The intersection of this line with the curved surface is the pressure center.



Example 1 a semi-cylinder of width $b=1\text{m}$.

Can it rotate endlessly under the action of F ?



Solution: 1, Conceptual analysis: It cannot
because the line of action of the total
pressure goes through the axis.

2, Calculation:

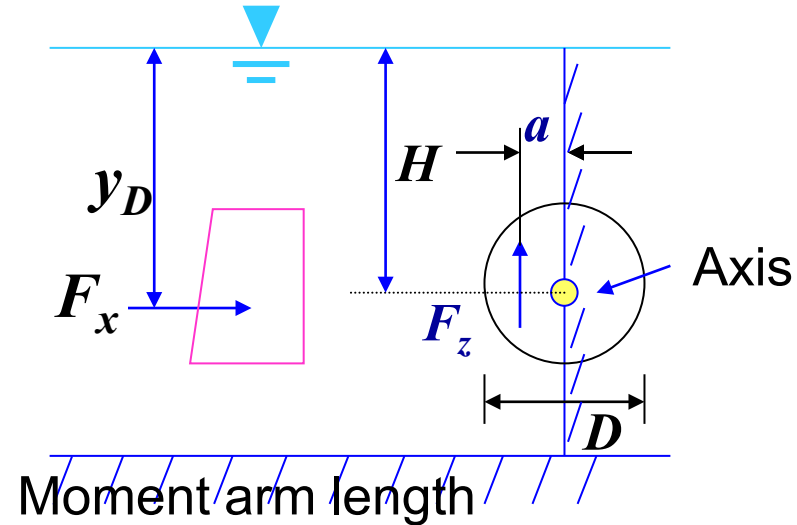
$$F_z = \rho g V = \rho g \times \frac{\pi}{4} D^2 \times \frac{1}{2} \times 1 = \frac{\pi}{8} D^2 \rho g$$

$$F_x = \rho g h_c A = \rho g \times H \times D \times 1 = \rho g H D$$

$$y_D = y_c + \frac{I_{cx}}{y_c A}$$

$$y_D - y_c = \frac{I_{cx}}{y_c A} = \frac{\frac{1}{12} \times 1 \times D^3}{H \times D \times 1} = \frac{D^2}{12H}$$

$$\therefore M_x = -F_x \cdot \frac{D^2}{12H} = -\frac{\rho g D^3}{12}$$

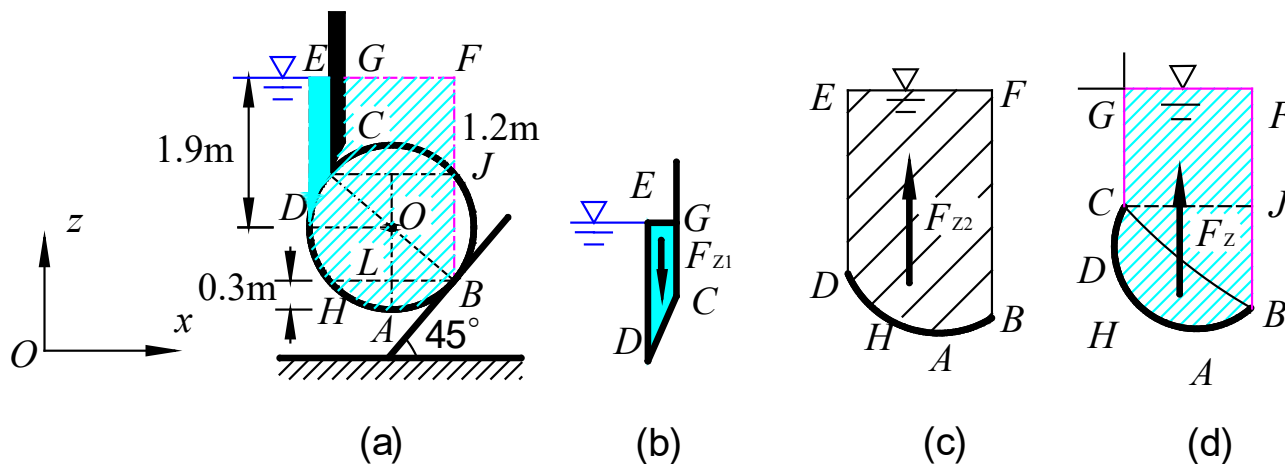


$$a = \frac{2D}{3\pi}$$

$$\therefore M_z = F_z \cdot a = \frac{\pi}{8} D^2 \rho g \times \frac{2D}{3\pi} = \frac{\rho g D^3}{12}$$

$$\therefore \sum M = M_x + M_z = 0$$

Example 2 Diameter of cylinder is **2m**, other dimensions, see Figure (a). There is water on the left and no water on the right. Find F_x and F_z of acting on the cylinder per meter of length.



$$F_x = \rho g h_c A = 9800 \times (1.2 + 0.7) \times 1.4 \times 1 = 26068 \text{ N}$$

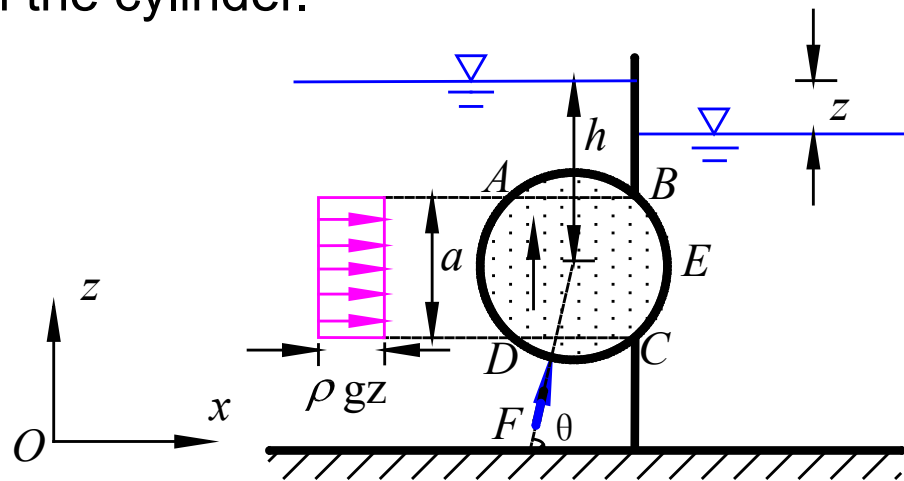
$F_z = (\text{rectangle } JFGC + \text{triangle } CJB + \text{semicircle } CDHAB) * \text{water weight}$

$$\begin{aligned} F_z &= 9800 \times (1.2 \times 1.4 + \frac{1}{2} 1.4 \times 1.4 + \frac{1}{2} \pi \times 1^2) \times 1 \\ &= 9800 \times (1.68 + 0.98 + 1.57) = 41454 \text{ N} \end{aligned}$$



Example 3 a rectangular hole on a vertical wall: height $a=1.0\text{m}$, width $b=3\text{m}$. A cylindrical cylinder of diameter $d=2\text{m}$ blocks it. Both sides of the partition are filled with water, $h=2\text{m}$, $z=0.6\text{m}$. Find the total hydrostatic force acting on the cylinder.

$$F_x = abz\rho g = 1.0 \times 3 \times 0.6 \times 9.806 \\ = 17.65 \text{ kN}$$



$$F_z = \rho g V_p = \rho g \left(\frac{\pi}{4} d^2 \times b \right) = 9.806 \times \left(\frac{\pi}{4} \times 2^2 \times 3 \right) = 92.37 \text{ kN}$$

$$F = \sqrt{F_x^2 + F_z^2} = \sqrt{17.65^2 + 92.37^2} = 94.04 \text{ kN}$$

$$\theta = \arctan \frac{F_z}{F_x} = \arctan \frac{92.37}{17.65} = 79.18^\circ$$

$$h_D = h + \frac{d}{2} \sin \theta = 2 + \frac{2}{2} \sin 79.18^\circ = 2.98 \text{ m}$$



Summary

1, **Calculation of horizontal force**: The vertical projection plane on the curved surface is the force plane, and the horizontal force is calculated from the analytical method of the total hydrostatic force on a plane;

$$F_x = \rho g h_c A_z$$

2, **Calculation the vertical force**: Draw the pressure body of the surface. The volume of the pressure body is bounded by the following surfaces: the surface under pressure itself, the vertical surface drawn through the surrounding edges of the surface, the liquid surface or an extension of the liquid surface. The magnitude of the vertical component force is the weight of the pressure body.

$$F_z = \rho g V_p$$

$$F = \sqrt{F_x^2 + F_z^2}$$

$$\theta = \arctan \frac{F_z}{F_x}$$

